Trainings

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Depending on the circumstance, diesel fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

Note : If an air compressor is **not** mounted on the engine, it will **not** be necessary to drain the cooling system.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Remove the fuel pump. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html)
- Remove the air compressor. Refer to Procedure 012-014 in Section 12. (/qs3/pubsys2/xml/en/procedures/56/56-012-014-tr.html)

- Remove the accessory drive pulley. Refer to Procedure 009-004 in Section 9. (/qs3/pubsys2/xml/en/procedures/56/56-009-004-tr.html)
- Remove the accessory drive cover, if applicable. Refer to Procedure 009-039 in Section 9. (/qs3/pubsys2/xml/en/procedures/56/56-009-039-tr.html)
- Remove the accessory drive seal, if applicable. Refer to Procedure 001-003 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-003-tr.html)

with Electronically Actuated Injector

⚠ WARNING ⚠

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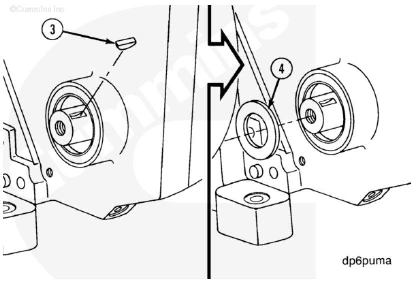
- Remove the fuel pump. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html)
- Remove the accessory drive pulley. Refer to Procedure 009-004 in Section 9. (/qs3/pubsys2/xml/en/procedures/56/56-009-004-tr.html)
- Remove the accessory drive cover, if applicable. Refer to Procedure 009-039 in Section 9. (/qs3/pubsys2/xml/en/procedures/56/56-009-039-tr.html)
- Remove the accessory drive seal, if applicable. Refer to Procedure 001-003 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-003-tr.html)

Remove

with Mechanically Actuated Injector

⚠ CAUTION ⚠

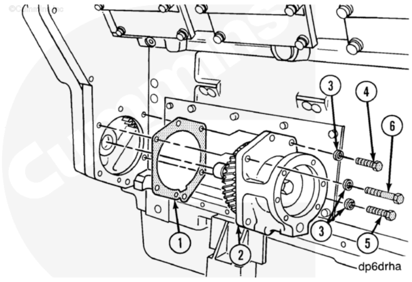
Make sure the Woodruff key has been removed before attempting to remove the fuel pump drive assembly. Damage to the bushing in the front gear housing cover can result if the key is not removed.



Remove the oil slinger.

Remove the six capscrews from the drive housing flange.

Remove the drive and discard the gasket.

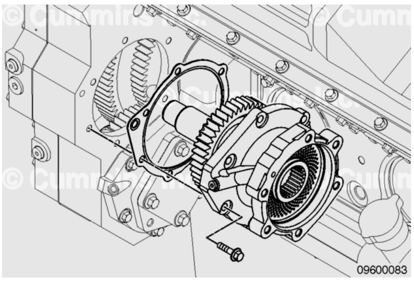


with Electronically Actuated Injector

Remove the six capscrews.

Remove the accessory drive assembly.

Remove and inspect the gasket.

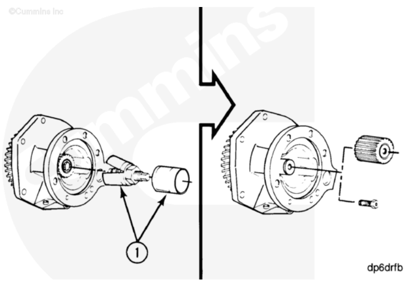


Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

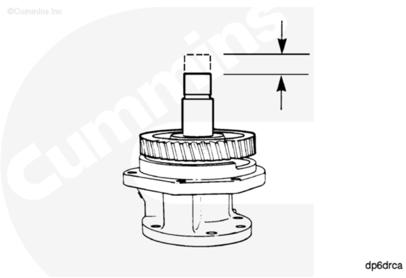
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Use solvent to clean the outer housing of the accessory drive.

Use a dial indicator to measure the accessory driveshaft end clearance.

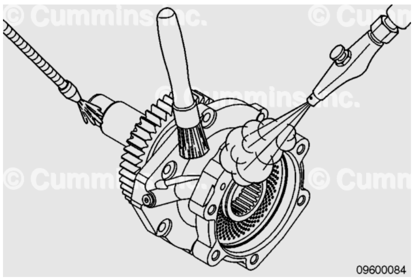
Accessory Driveshaft End Clearance		
mm		in
0.05	MIN	0.0020
0.30	MAX	0.0118



with Electronically Actuated Injector

⚠ WARNING ⚠

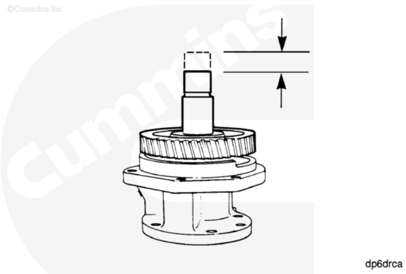
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Use solvent to clean the outer housing of the accessory drive.

Use a dial indicator to measure the fuel pump driveshaft end clearance.

Fuel Pump Driveshaft End Clearance		
mm		in
0.13	MIN	0.0051
0.29	MAX	0.0114

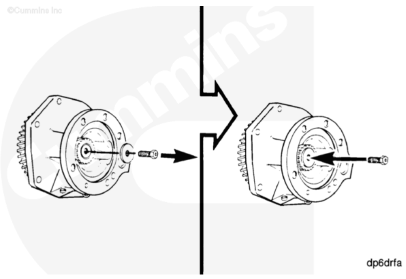


Disassemble

with Mechanically Actuated Injector

⚠ CAUTION ⚠

To prevent damage to the shaft, the capscrew must be installed into the drive shaft without the washer.

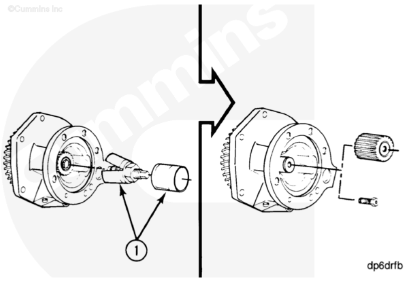


Remove the capscrew and washer.

Install the capscrew into the shaft.

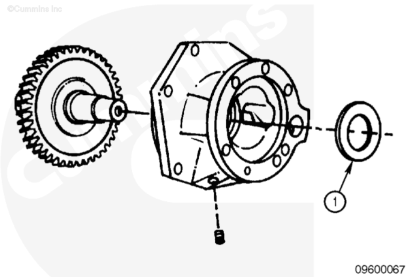
Use a jaw puller to remove the Lovejoy or spline type coupling.

Remove the capscrew from the shaft.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Remove the clamping washer (1).

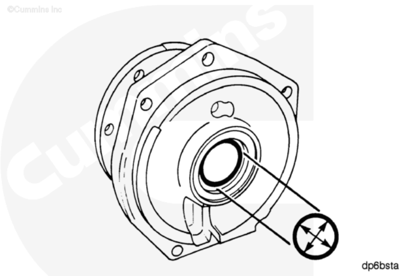
Remove the gear and shaft assembly.

Remove the pipe plug from the housing.

Use solvent to clean the parts.

Measure the inside diameter of the bushings.

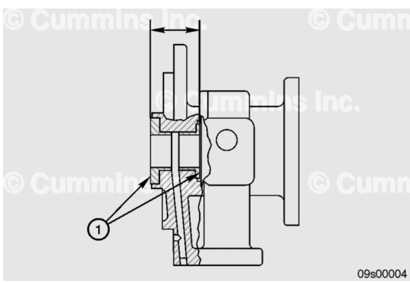
Accessory Drive Bushing Inside Diameter		
mm		in
49.95	MIN	1.9665
50.03	MAX	1.9697



If the inside diameter of either bushing is **not** within specification, both bushings **must** be replaced.

Check the grooved surfaces of the bushings (1) for damage.

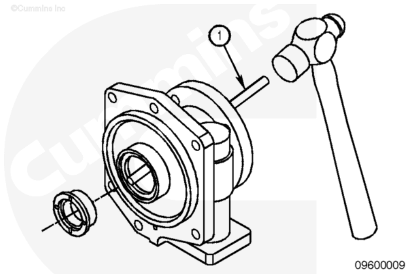
Measure the distance from the outside of the thrust faces of the two bushings.



Accessory Drive Bushing Thrust Face Distance		
mm		in
45.37	MIN	1.7862
46.45	MAX	1.8287

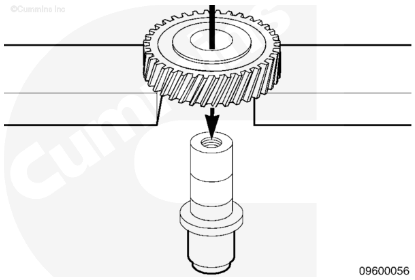
If the distance between the two thrust faces is **not** within specification, both bushings **must** be replaced.

Remove the flanged bushing from the drive housing using a brass drift (1), taking care **not** to damage the bushing bore.



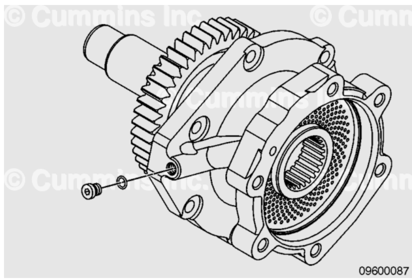
Only remove the gear from the shaft when the gear or the shaft **must** be replaced.

Use an arbor press to remove the gear.

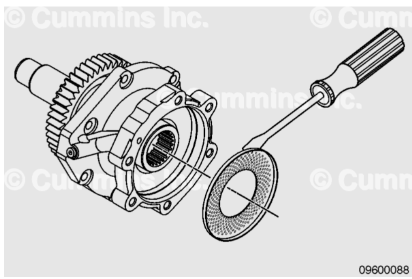


with Electronically Actuated Injector

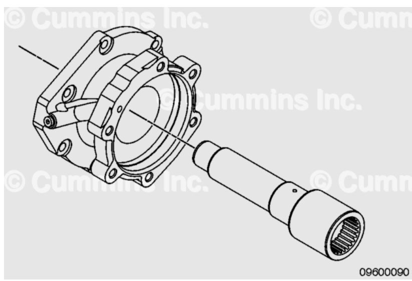
Remove the straight thread o-ring plug from the fuel pump drive.



Remove the mesh screen with a small flat tip screwdriver.

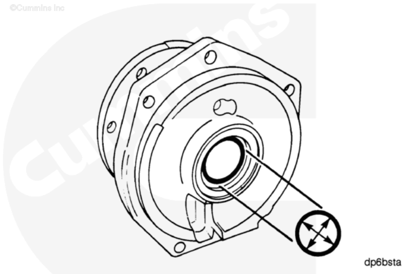


Use a press to remove the shaft from both the gear and the pump drive housing.



Measure the inside diameter of the bushings.

Fuel Pump Drive Bushing Inside Diameter		
mm		in
33.57	MIN	1.3172
33.64	MAX	1.3244

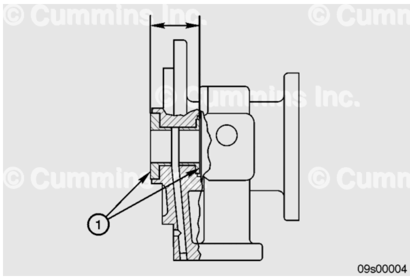


If the inside diameter of either bushing is **not** within specification, both bushings **must** be replaced.

Check the grooved surfaces of the bushings (1) for damage.

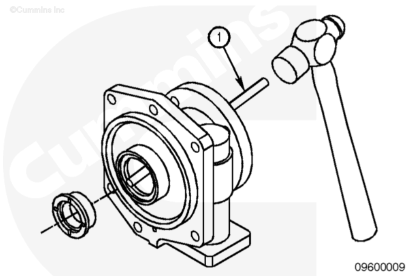
Measure the distance from the outside of the thrust faces of the two bushings.

Fuel Pump Drive Bushing Thrust Face Distance		
mm		in
45.52	MIN	1.7921
45.64	MAX	1.7968



If the distance between the two thrust faces is **not** within specification, both bushings **must** be replaced.

Remove the flanged bushing from the drive housing. Use a brass drift (1), taking care **not** to damage the bushing bore.



Clean and Inspect for Reuse

with Mechanically Actuated Injector

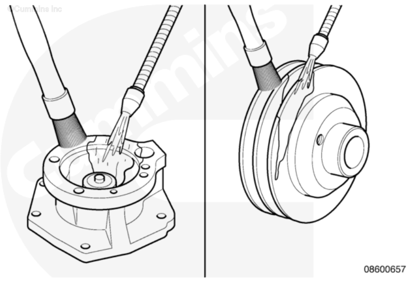
⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of

personal injury.

⚠ WARNING ⚠

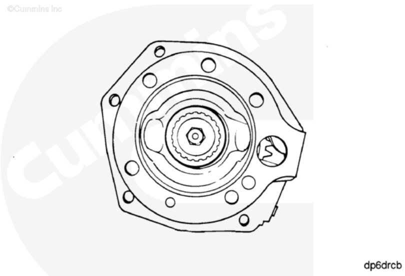
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Clean the fuel pump drive assembly and inspect it for reuse.

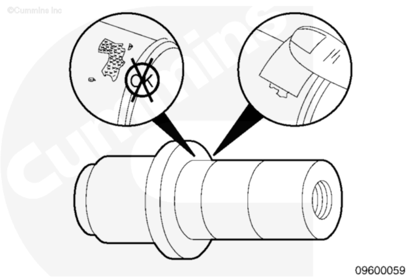
Clean the pulley and inspect if for reuse, if applicable.

The coupling washer **must** be clamped tightly between the coupling and the shaft. If the washer is **not** tight, rebuild the drive unit.



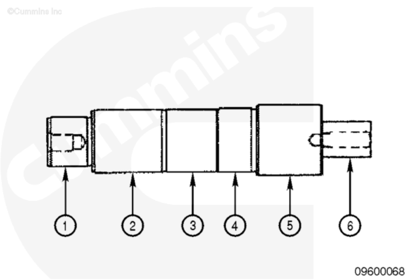
Inspect the shaft in the gear fit area for fretting or burrs.

Clean burrs with emery cloth.



Measure the shaft outside diameter.

Accessory Driveshaft Outside Diameter			
	mm		in
(1)	34.964	MIN	1.3765
	34.976	MAX	1.3770
(2)	43.928	MIN	1.7294
	43.940	MAX	1.7299
(3)	44.370	MIN	1.7468
	44.630	MAX	1.7571
(4)	45.070	MIN	1.7744
	45.082	MAX	1.7749
(5)	49.820	MIN	1.9614
	49.832	MAX	1.9619

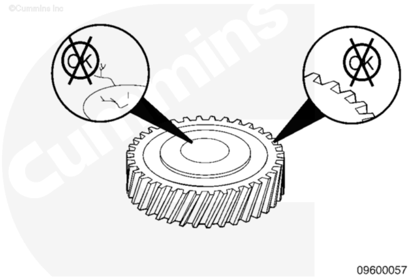


(6)	25.325	MIN	0.9970
	25.375	MAX	0.9990

If the outside diameter is **not** within specification, the shaft **must** be replaced.

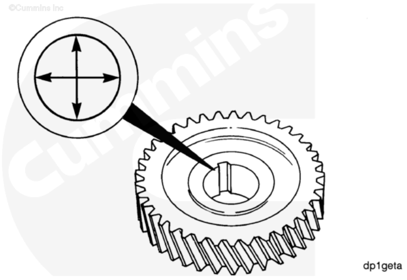
Check the gear teeth and gear bore for damage.

Replace the gear, if damaged.



Measure the gear inside diameter.

Accessory Drive Gear Inside Diameter		
mm		in
44.98	MIN	1.7709
45.02	MAX	1.7724

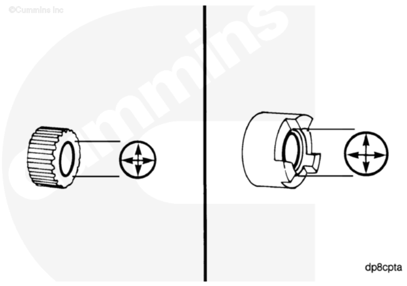


If the gear is **not** within specification, the gear **must** be replaced.

Check the inner diameter of the coupling for damage.

If damage is found, the coupling **must** be replaced.

Measure the inside diameter of the coupling.



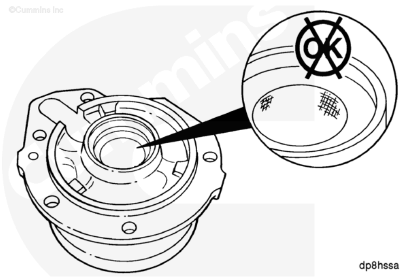
Fuel Pump Drive Lovejoy Coupling Inside Diameter		
mm		in
25.377	MIN	0.9991
25.407	MAX	1.0003

Fuel Pump Drive Spline Coupling Inside Diameter		
mm		in
25.255	MIN	0.9943
25.278	MAX	0.9952

If the inner diameter of the coupling is **not** within specification, the coupling **must** be replaced.

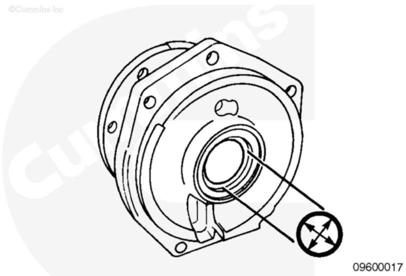
If the bushing was removed, inspect the housing for scoring and other damage.

Replace the housing if damaged.



If the bushing was removed, measure the inside diameter of the bushing bore.

Fuel Drive Drive Bushing Bore Inside Diameter		
mm		in
60.00	MIN	2.3622
60.05	MAX	2.3642



Replace the housing if the bushing bore is **not** within specifications.

with Electronically Actuated Injector

⚠ WARNING ⚠

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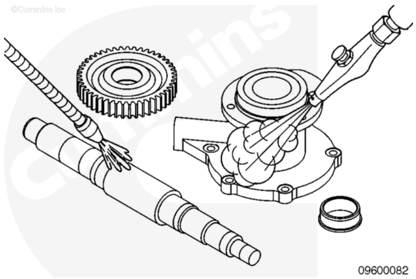


⚠ WARNING ⚠

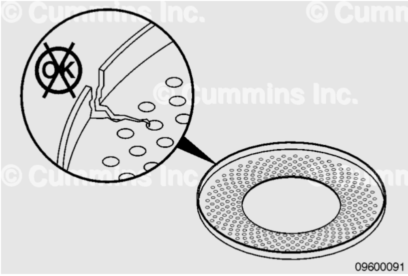
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the fuel pump drive components.

Dry with compressed air.

Inspect the screen for cracks or other damage.

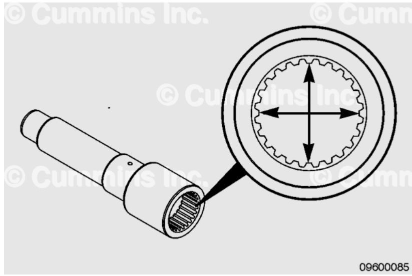
If any damage is found, the screen **must** be replaced.



Check the inner diameter of the shaft spline for damage.

If damage is found, the shaft **must** be replaced.

Measure the inside diameter of the shaft.

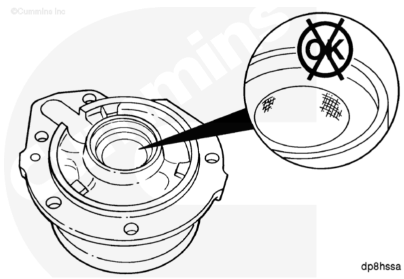


Fuel Pump Drive Spline Inside Diameter		
mm		in
46.000	MIN	1.8110
46.190	MAX	1.8185

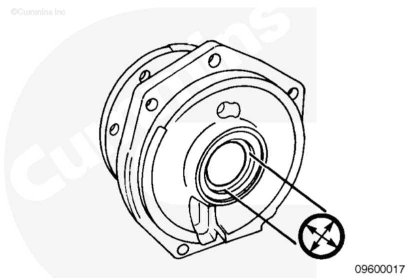
If the inner diameter of the shaft is **not** within specification, the shaft **must** be replaced.

If the bushing was removed, inspect the housing for scoring and other damage.

Replace the housing, if damaged.



If the bushing was removed, measure the inside diameter of the bushing bore.

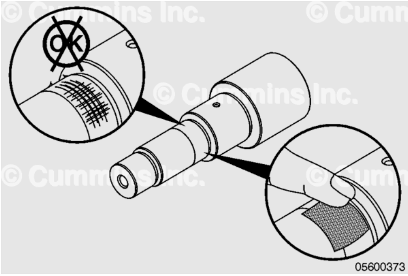


Fuel Pump Drive Bushing Bore Inside Diameter		
mm		in
40.26	MIN	1.5850
40.77	MAX	1.6051

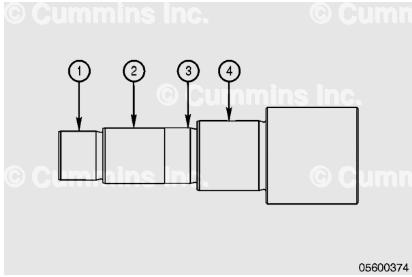
Replace the housing if the bushing bore is **not** within specifications.

Inspect the shaft in the gear fit area for fretting or burrs.

Clean burrs with emery cloth.



Measure the driveshaft outside diameter.



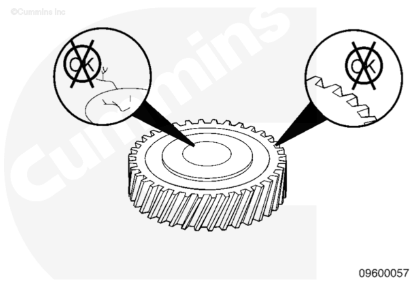
Fuel Pump Driveshaft Outside Diameter			
	mm		in
(1)	34.963	MIN	1.3765
	34.978	MAX	1.3771
(2)	43.928	MIN	1.7294
	43.940	MAX	1.7299
(3)	44.370	MIN	1.7468
	44.630	MAX	1.7571
(4)	49.820	MIN	1.9614
	49.832	MAX	1.9619

If the outside diameter is **not** within specification, the driveshaft **must** be replaced.

Note : Dimensions 2 and 3 are on the same section of the driveshaft. Dimension 3 only applies in the area adjacent to the shoulder, where the keyway is located.

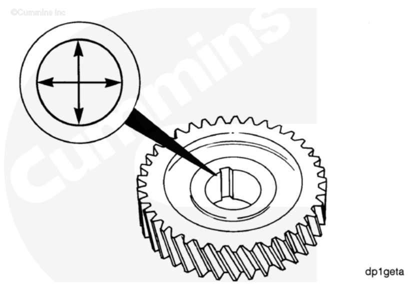
Check the gear teeth and gear bore for damage.

Replace the gear, if damaged.



Measure the gear inside diameter.

Fuel Pump Drive Gear Inside Diameter		
mm		in
39.73	MIN	1.5642
39.75	MAX	1.5650



If the gear is **not** within specification, the gear **must** be replaced.

Assemble

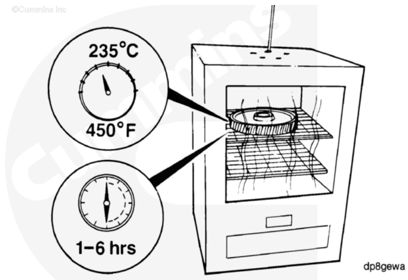
with Mechanically Actuated Injector

⚠ WARNING ⚠

Wear protective clothing to reduce the possibility of personal injury from burns.

⚠ CAUTION ⚠

Do not exceed the specified time or temperature. Damage to the gear teeth will result.

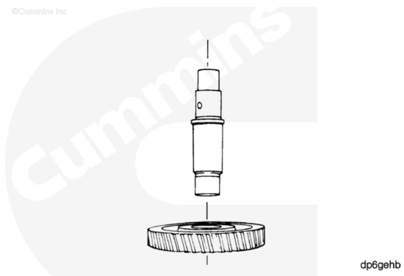


If an adequate press is **not** available, an oven can be used.

Heat the gear at 235°C [450°F] for **not** less than one hour, and **not** more than six hours.

⚠ WARNING ⚠

Wear protective clothing to reduce the possibility of personal injury from burns.



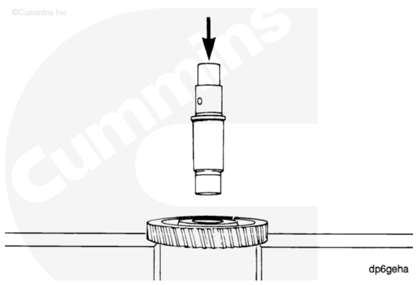
Support the gear.

Slide the shaft into the gear until the shoulder of the shaft touches the gear.

If an arbor press is available, support the gear.

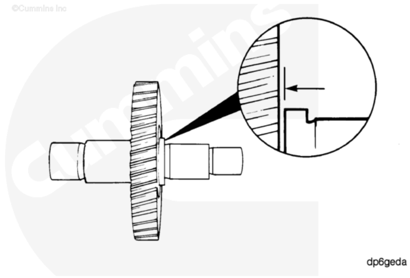
Lubricate the shaft with clean engine oil.

Press the shaft through the gear until the shoulder of the shaft touches the gear.



⚠ CAUTION ⚠

If the heat method was used to install the gear, allow the gear to air cool. Do not use water or oil to reduce cooling time. Damage to the gear can result.



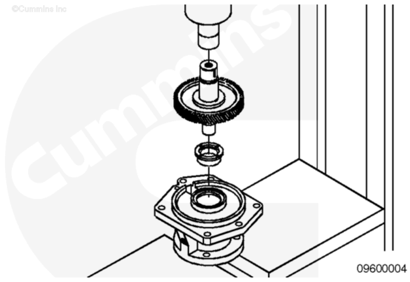
Use a feeler gauge to measure the distance between the shoulder of the shaft and the gear.

Accessory Drive Gear to Shaft Distance		
mm		in
0.05	MAX	0.0020

If the distance between the gear and the shaft is **not** within specification, press the gear on until the specification is met.

⚠ CAUTION ⚠

The bushings must be pressed in square to the bores of the housing. The thrust faces of the two bushings must be parallel and concentric. The bushing with the center counterbored must have the pressure applied to the counterbore face and not on the thrust bearing face. Failure to comply can result in distortion of the thrust face.



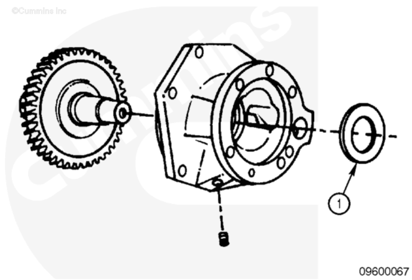
Support the housing.

The bushing with the smaller outside diameter is pressed into the output side.

Using an arbor press and the shaft as a mandrel, press the bushing into the housing.

Turn the housing over and press the remaining bushing into place.

Lubricate the grooved surface of the thrust bearing with Lubriplate™ 105, Part Number 3163086, 3163087, or equivalent.



Install the gear and shaft assembly into the housing.

Install the clamping washer (1) with the beveled edge up.

Install the pipe plug into the housing.

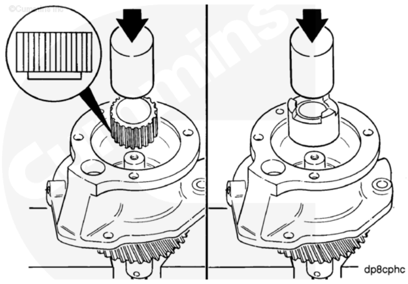
Tighten the pipe plug.

Torque Value: 8 n•m [71 in-lb]

Support the assembly on the gear or the shaft.

Use an arbor press and a mandrel to press the coupling onto the shaft until it touches the clamping washer.

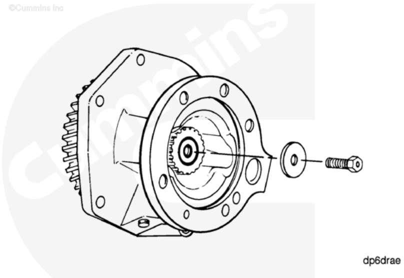
The clamping washer **must** be positioned tightly between the coupling and the shoulder of the shaft.



The capscrew **must** contain an oil drilling if an air compressor is to be mounted on the engine.

Install the washer and capscrew.

Tighten the capscrew.

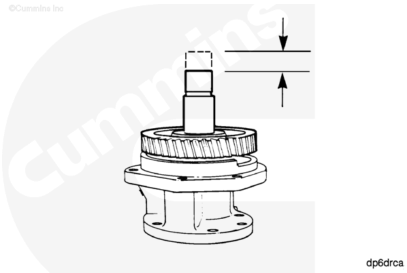


Torque Value:
9.525 mm [0.375 in] Capscrew Length 45 n•m [33 ft-lb]

Torque Value:
12.7 mm [0.50 in] Capscrew Length 100 n•m [74 ft-lb]

Rotate the driveshaft, checking for correct assembly.

Use a dial indicator to measure the accessory driveshaft end clearance.

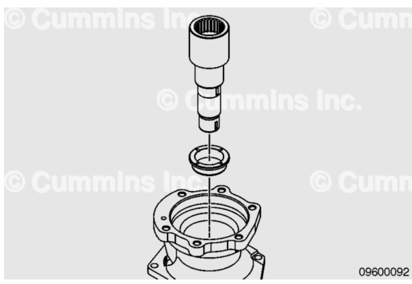


Accessory Driveshaft End Clearance		
mm		in
0.05	MIN	0.0020
0.30	MAX	0.0118

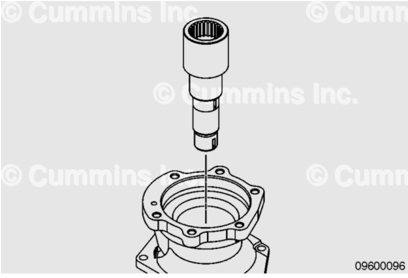
If the end clearance is **not** within specifications, make sure the coupling is positioned tightly against the clamping washer.

with Electronically Actuated Injector

If the bushings were removed, use the shaft as a mandrel and press one bushing into the housing until the underside of the bushing head is flush with the counterbore in the housing. Turn the housing over and press the remaining bushing into place.



Install the shaft into the housing.



⚠ WARNING ⚠

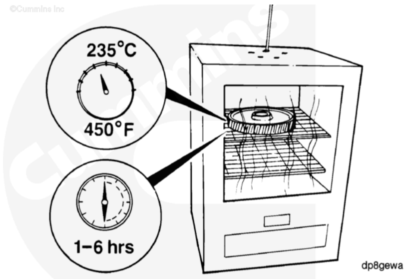
Wear protective clothing to reduce the possibility of personal injury from burns.

⚠ CAUTION ⚠

Do not exceed the specified time or temperature. Damage to the gear teeth will result.

If an adequate press is **not** available, an oven can be used.

Heat the gear to 204°C [400°F] for **not** less than 1 hour, and **not** more than 6 hours.



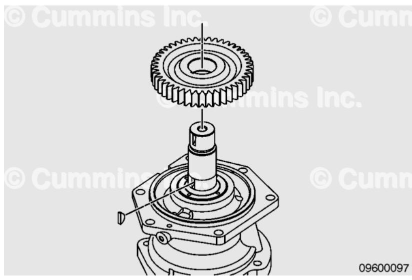
⚠ WARNING ⚠

Wear protective clothing to reduce the possibility of personal injury from burns.

Install the key onto the shaft.

Support the shaft and housing.

Slide the gear onto the shaft until the shoulder of the shaft touches the gear.

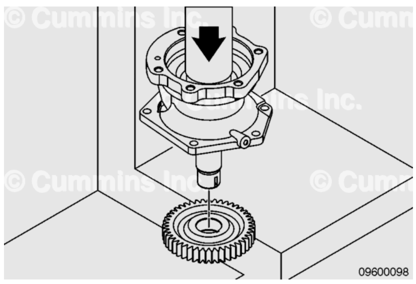


If an arbor press is available, support the gear.

Install the key onto the shaft.

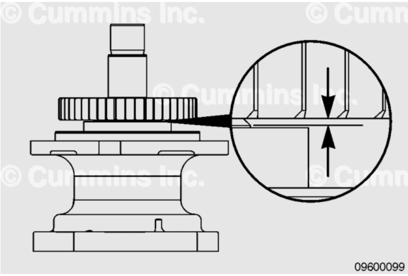
Lubricate the shaft with clean engine oil.

Press the shaft through the gear until the shoulder of the shaft touches the gear.



⚠ CAUTION ⚠

If the heat method was used to install the gear, allow the gear to air cool. Do not use water or oil to reduce cooling time. Damage to the gear can result.

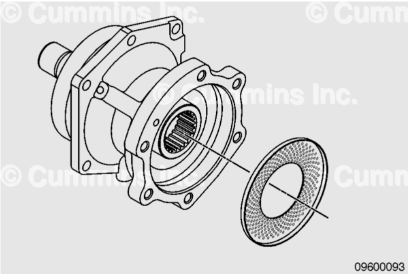


Use a feeler gauge to measure the distance between the shoulder of the shaft and the gear.

Accessory Drive Gear to Shaft Distance		
mm		in
0.05	MAX	0.0020

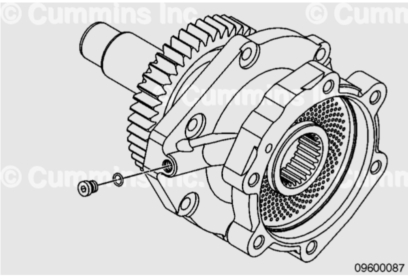
If the distance between the gear and the shaft is **not** within specification, press the gear on until the specification is met.

Install the screen into the housing by pushing the screen in by hand.



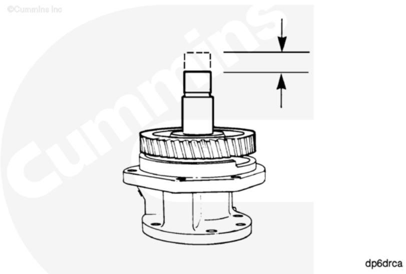
Install the straight thread o-ring plug.

Torque Value: 6 n•m [53 in-lb]



Rotate the driveshaft, checking for correct assembly.

Use a dial indicator to measure the fuel pump driveshaft end clearance.



Fuel Pump Driveshaft End Clearance		
mm		in
0.13	MIN	0.0051
0.29	MAX	0.0114

If the end clearance is **not** within specifications, make sure the coupling is positioned tightly against the clamping washer.

Install

with Mechanically Actuated Injector

Note : This procedure also applies to the drive installed on engines that have a governor drive.

Use Lubriplate™ to lubricate the bushing in the front gear housing cover.

Note : Do not use sealants on the gasket. The gasket is manufactured from the material that sealants will damage.

Install the gasket (1) onto the pilot of the housing cover.

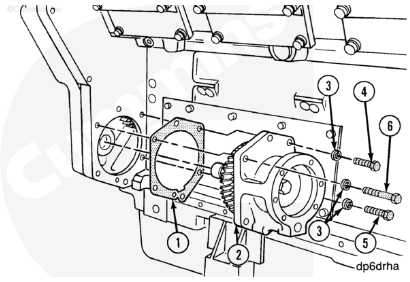
Install the drive assembly (2), turning the shaft as necessary to engage the gears.

Capscrew Specifications:

1. M10 X 1.50 X 35

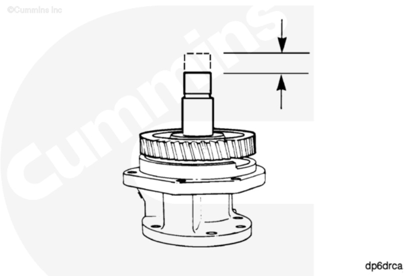
Tighten the capscrews alternately and evenly.

Torque Value: 45 n•m [33 ft-lb]

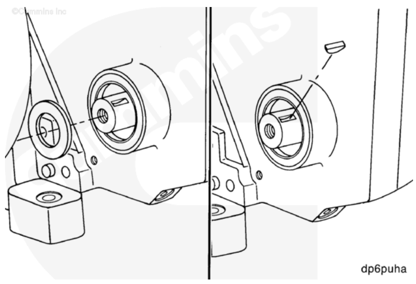


With the accessory drive installed, measure the end clearance.

Fuel Pump Drive End Clearance		
mm		in
0.05	MIN	0.0020
0.30	MAX	0.0118



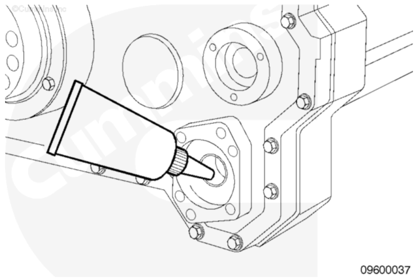
Install the oil slinger.



with Electronically Actuated Injector

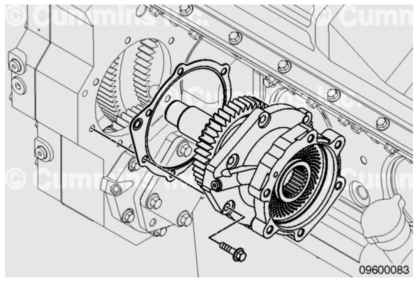
⚠ CAUTION ⚠

Do not use sealants on the gasket. The gasket is manufactured from material that sealants will damage.



Lubricate the bushing in the front gear housing cover with Lubriplate™ 105.

Install the gasket onto the pilot of the drive housing cover.

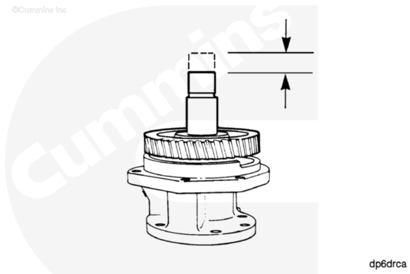


There is **not** a requirement to align the index on the drive gear with the marks on the camshaft gear on QSK60 engines.

Install the fuel pump drive assembly, turning the shaft as necessary to engage the gears.

Install and tighten the six capscrews.

Torque Value: 45 n•m [33 ft-lb]



With the fuel pump drive installed, measure the end clearance.

Fuel Pump Drive End Clearance		
mm		in
0.13	MIN	0.0051
0.29	MAX	0.0114

Finishing Steps

with Mechanically Actuated Injector

- Install the accessory drive seal, if applicable. Refer to Procedure 001-003 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-003-tr.html)
- Install the accessory drive pulley, if applicable. Refer to Procedure 009-004 in Section 9. (/qs3/pubsys2/xml/en/procedures/56/56-009-004-tr.html)
- Install the accessory drive cover, if applicable. Refer to Procedure 009-039 in Section 9. (/qs3/pubsys2/xml/en/procedures/56/56-009-039-tr.html)
- Install the air compressor. Refer to Procedure 012-014 in Section 12. (/qs3/pubsys2/xml/en/procedures/56/56-

012-014-tr.html)
- Install the fuel pump. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Operate the engine until the coolant temperature reaches 71°C [160°F]. Check for coolant leaks.

with Electronically Actuated Injector

- Install the accessory drive seal, if applicable. Refer to Procedure 001-003 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-003-tr.html)
- Install the accessory drive pulley, if applicable. Refer to Procedure 009-004 in Section 9. (/qs3/pubsys2/xml/en/procedures/56/56-009-004-tr.html)
- Install the accessory drive cover, if applicable. Refer to Procedure 009-039 in Section 9. (/qs3/pubsys2/xml/en/procedures/56/56-009-039-tr.html)
- Install the fuel pump. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html)
- Operate the engine until the coolant temperature reaches 71°C [160°F] and check for leaks.

Last Modified: 16-Oct-2023